

1. Suppose the employee sends an HTTP/HTTPS request to a web server whose public IP address is **203.0.113.10**. Describe, step by step, what changes the branch router makes to the outgoing packet before forwarding it to the Internet. Clearly mention the role of **source IP address, public IP address, source port, destination IP address, and NAT translation table**.
2. Suppose the client chooses an initial sequence number **1000**, while the server chooses an initial sequence number **5000**. Show the **SYN, SYN-ACK, and ACK** exchange and determine the sequence and acknowledgment numbers used in each segment.
3. The DNS resolver normally uses **UDP** for ordinary DNS queries. Explain the technical reasons behind this choice. Under what circumstances might **TCP** be used instead?
4. Suppose the DNS server returns the IP address **203.0.113.50** for `www.bankexample.com`. Explain what happens **after the browser receives this IP address** and how the computer uses it to communicate with the web server.
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 - (a) Explain why knowing the destination **IP address alone is not sufficient** for the sender to deliver an Ethernet frame within the local network. What additional information is required?
 - (b) Describe the **step-by-step ARP process** that occurs when 192.168.1.10 wants to communicate with 192.168.1.20.

6.Scenario:

A bank has two departments, **Accounts** and **IT**, located on the same floor. Each department has several computers. The bank also has another branch located in a different city, and both branches need to communicate with each other and access the Internet.

Answer the following questions:

- (a) The Accounts department has 10 computers that need to communicate efficiently within the same LAN. Which networking device would be more appropriate: **Hub or Switch**? Explain your reasoning based on how each device handles incoming frames.
- (b) Suppose two separate LAN segments need to be connected within the same organization. Explain how a **Bridge** can help connect these networks. How is its operation different from simply broadcasting every frame to all connected devices?
- (c) The bank's branch network needs to communicate with a network located in another city. Which device is primarily responsible for forwarding packets between **different IP networks**? Explain how it makes its forwarding decision.
- (d) An employee's computer wants to access a server located outside its own local network. Explain the role of the **default gateway** in this communication. Is a gateway simply another name for a router? Explain the relationship between the two terms.